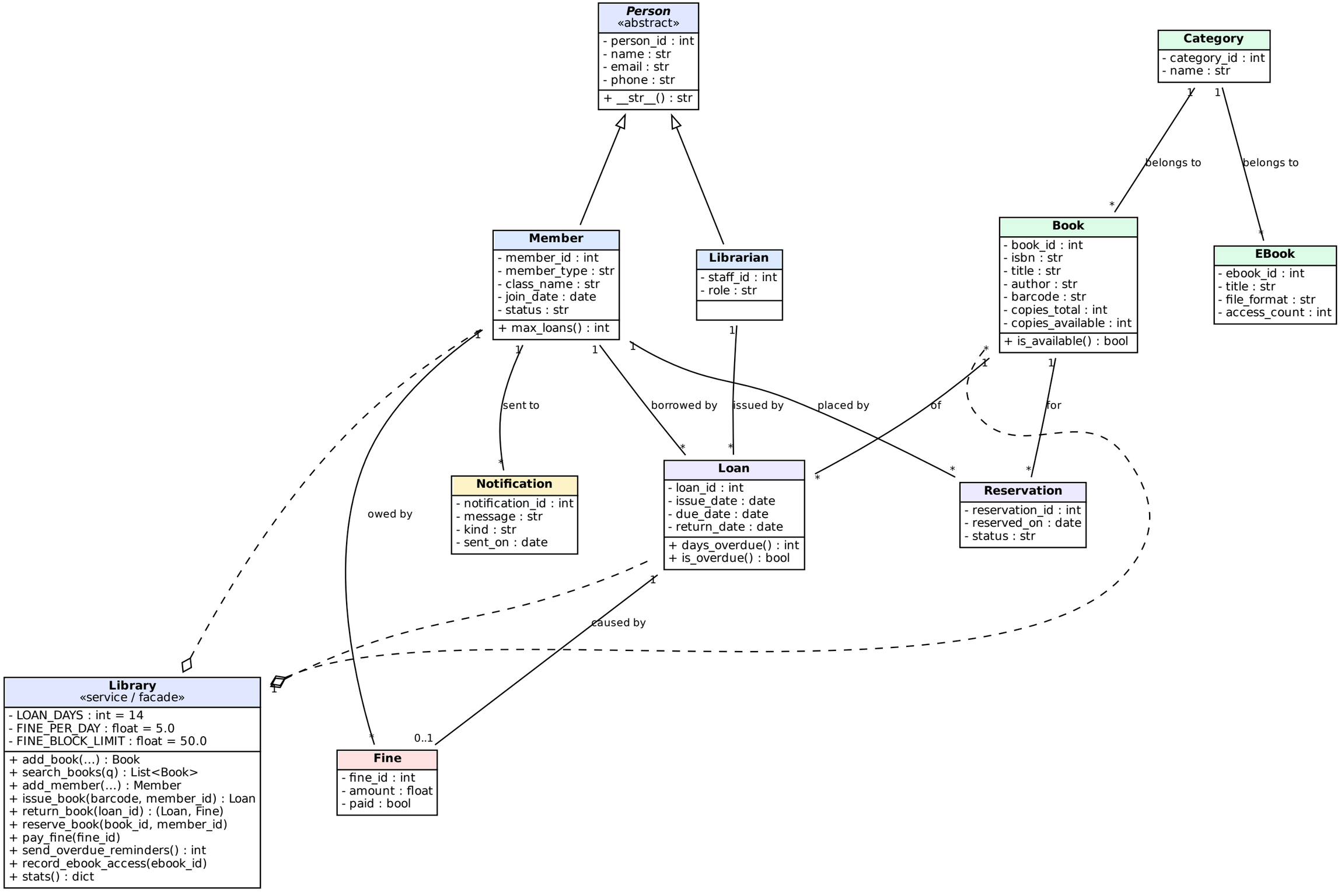
Person
«abstract»
- person_id : int
- name : str
- email : str
- phone : str
+ __str__() : str
Category
- category_id : int
- name : str
belongs to
belongs to
Member
- member_id : int
- member_type : str
- class_name : str
- join_date : date
- status : str
+ max_loans() : int
Librarian
- staff_id : int
- role : str
Book
- book_id : int
- isbn : str
- title : str
- author : str
- barcode : str
- copies_total : int
- copies_available : int
+ is_available() : bool
EBook
- ebook_id : int
- title : str
- file_format : str
- access_count : int
sent to
borrowed by
issued by
placed by
of
for
owed by
caused by
Notification
- notification_id : int
- message : str
- kind : str
- sent_on : date
Loan
- loan_id : int
- issue_date : date
- due_date : date
- return_date : date
+ days_overdue() : int
+ is_overdue() : bool
Reservation
- reservation_id : int
- reserved_on : date
- status : str
Library
«service / facade»
- LOAN_DAYS : int = 14
- FINE_PER_DAY : float = 5.0
- FINE_BLOCK_LIMIT : float = 50.0
+ add_book(...) : Book
+ search_books(q) : List<Book>
+ add_member(...) : Member
+ issue_book(barcode, member_id) : Loan
+ return_book(loan_id) : (Loan, Fine)
+ reserve_book(book_id, member_id)
+ pay_fine(fine_id)
+ send_overdue_reminders() : int
+ record_ebook_access(ebook_id)
+ stats() : dict
Fine
- fine_id : int
- amount : float
- paid : bool